



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

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Comparison Table

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Paid, per-core licensing (Standard/Datacenter editions); evaluation is free for 180 days	Free and open-source (Canonical offers paid support/Ubuntu Pro optionally)	Free and open-source, no paid tier required (community-driven, RHEL-compatible)
User Interface	Offers both GUI (Desktop Experience) and Server Core (CLI-only); GUI is user-friendly for Windows admins	CLI-only by default (no GUI on server edition); managed via terminal/SSH	CLI-only by default, similar to other RHEL-based distros; managed via terminal/SSH
Package Management	Uses PowerShell, Windows Update, and MSI/EXE installers; also supports winget	Uses APT (apt install, apt update) — Debian-based	Uses DNF/YUM (dnf install) — Red Hat-based
Security	Built-in Windows Defender, Active Directory, Group Policy; frequent patches via Windows Update	AppArmor for access control, UFW firewall, regular security patches via APT	SELinux (stricter mandatory access control by default), firewalld, known for enterprise-grade hardening
Performance	Heavier resource usage (especially with GUI); needs more RAM/CPU to run comfortably	Lightweight, efficient on lower-spec hardware; great for VMs and cloud instances	Lightweight and stable, performance comparable to Ubuntu Server, tuned for long-term enterprise stability
Typical Enterprise Use Cases	Active Directory domains, .NET applications, SQL Server, enterprises already invested in Microsoft ecosystem	Web hosting, cloud infrastructure (AWS/Azure/GCP), Docker/Kubernetes, DevOps pipelines	Enterprise servers needing RHEL compatibility without licensing cost (banking, government, data centers)
Advantages	Deep integration with Microsoft products, strong enterprise support contracts, familiar GUI option	Huge community support, frequent updates, most popular cloud/server Linux distro, free	Very stable, long support lifecycle (10 years), binary-compatible with RHEL, free alternative to CentOS
Disadvantages	Costly licensing, higher resource requirements, more	Less enterprise "one-stop" support compared to	Smaller community than Ubuntu, less beginner



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	vulnerable attack surface with GUI enabled	RHEL/Windows without a paid plan	documentation, newer project (created after CentOS discontinued)
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Discussion

Windows Server, Ubuntu Server, and Rocky Linux all do similar jobs but they're really built for different situations. Windows Server makes the most sense if a company is already using Microsoft stuff — Active Directory, .NET apps, SQL Server, all of that connects easily since it's made by the same company. It also has a GUI option (Desktop Experience) which makes it way easier for beginners or people used to regular Windows. The downside is it costs money — the licensing is per-core, so it adds up fast for bigger setups, and if you turn on the GUI it eats up a lot more RAM and CPU than it really needs to.

Ubuntu Server is the opposite in a lot of ways. It's completely free, lightweight, and probably the most common Linux distro you'll see running on cloud servers right now. I actually installed this one for this project, and honestly the installer was pretty simple to get through once I figured out the navigation (Tab/Enter instead of a mouse). It didn't need much RAM either, which mattered a lot since my laptop only has 3.5GB total. It uses APT for installing software (apt install, apt update) which is pretty beginner-friendly too, and there's a ton of documentation online since so many people use it.

Rocky Linux is less well-known but it exists because CentOS (which used to be the free version of RHEL) got discontinued by Red Hat. So Rocky Linux basically stepped in to be a free, community-run alternative that's still compatible with RHEL. Companies that need something stable and RHEL-compatible but don't want to pay for RHEL itself would go with Rocky. It uses SELinux for security, which is stricter than what Ubuntu uses (AppArmor), but that also makes it a bit harder to learn if you're not used to it.

Basically, it comes down to what the company already uses and how much they're willing to spend. If they're a Microsoft shop, Windows Server makes sense despite the cost. If they want something free and easy for cloud/DevOps stuff, Ubuntu is the go-to. And if they need RHEL-level stability without paying for RHEL, Rocky Linux is the pick.